GUILHERME HENRIQUE RISSON SILVA

Software Engineer Student | Backend & Data | Python | PostgreSQL | FastAPI | LLM
Maringá, Brazil (remote, hybrid or on-site) • +55 47 98911-5314 • guilhermerisson@icloud.com
linkedin.com/in/guilhermerisson • github.com/guilhermerissondev • guilhermerissondev.github.io

SUMMARY

Software Engineering student at Uninter and data analyst at Vitru Educação, building the Python systems behind UniCesumar's academic planning — especially the faculty allocation optimizer (OR-Tools, FastAPI, React). Open-source highlight: Job Hunters, a full-stack SaaS (FastAPI, Next.js, PostgreSQL/pgvector, LLM) that collects ~98k jobs from 649 sources and recommends applications with CV evidence; plus agents and RAG with verifiable citations.

EXPERIENCE

Vitru Educação (UniCesumar) 03/2026 – Present
Mid-level Data Analyst, Academic Planning Hybrid, Maringá, Brazil

- Built the faculty allocation optimizer (Python, OR-Tools CP-SAT, FastAPI, React, TypeScript), which assigns about 200 instructors to more than 350 classes per module in minutes, work that used to take weeks in spreadsheets.
- Built the 2027 student enrollment forecasting model (Python, DuckDB, Parquet, Power BI), with 50 automated quality checks and error below 2% against actuals.
- Maintain in production the weekly curriculum pipeline for UniCesumar and Uniasselvi (Python, SQL Server), which reads 6 spreadsheet layouts through 4 validation layers.
- Moved the planned-vs-actual audits from Power Query to Python, over datasets of up to 40 million rows, with Power BI dashboards for managers and leadership.
- Rewrote in Python and Marimo the lab class demand vs. tutor capacity model, with a simulator by campus and city whose scenarios cut the shortfall from 27,400 to 21,300 hours.

Linoforte Móveis 06/2022 – 12/2025
Assistant Operator Osvaldo Cruz, Brazil

- Set up monitoring of production and quality KPIs, flagging recurring stoppages for corrective action.
- Mapped bottlenecks and downtime and proposed improvements that reduced rework and waste on the line.

PROJECTS (github.com/guilhermerissondev)

- Job Hunters: job-hunting SaaS — FastAPI, Next.js, PostgreSQL/pgvector and LLM with verified citations; collects ~98k jobs from 649 sources and sends Telegram matches with CV evidence.
- Teacher Allocation Optimizer: faculty allocation with CP-SAT, FastAPI API and React dashboard (open-source of the system used at Vitru), with validation, audit and scenario simulation.
- TikTok Viral Generator: agent that tracks 8 sources, researches, scripts and publishes short videos with a source for every fact (Python, Pydantic, SQLite). 4 videos/day, 691 tests.
- Regulatory RAG Eval: Q&A over 294 Brazilian Central Bank regulations with source citations (Python, BM25, embeddings, Gemini). Hybrid search raised MRR from 0.361 to 0.644.

EDUCATION

B.S. in Software Engineering — Uninter — 10/2026 – 2030 (in progress)
AI Software Engineer 4.0 Program (360 h) — Data Science Academy — 09/2026 – 12/2026 (expected)
Postgraduate Certificate in Data Science (360 h) — UniCesumar — 03/2026 – 10/2026
B.S. in Civil Engineering — FAI, Faculdades Adamantinenses Integradas — 2021 – 2025

SKILLS

Languages: Python, SQL, TypeScript
AI and LLM: RAG, hybrid search (BM25 + embeddings), structured output, LLM evaluation, agents, Gemini API
Backend and product: FastAPI, Next.js, React, Pydantic, PostgreSQL/pgvector, Docker, pytest, GitHub Actions, Git
Data and ML: pandas, DuckDB, Parquet, SQL Server, scikit-learn, LightGBM, OR-Tools (CP-SAT), Power BI
Spoken languages: Portuguese (native), English (professional), Spanish (intermediate)